

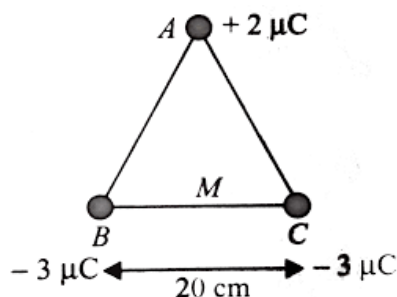
FREEDOM INTERNATIONAL SCHOOL

WORKSHEET

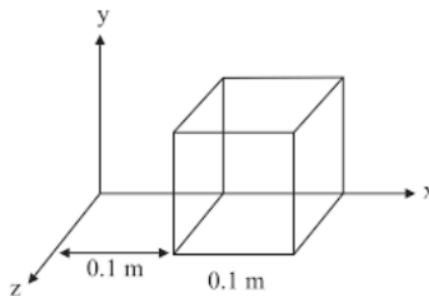
PHYSICS

ELECTRIC CHARGES AND FIELDS

1. Two charges of magnitudes $-2Q$ and $+Q$ are located at points $(a,0)$ and $(4a,0)$ respectively. What is the electric flux due to these charges through a sphere of radius $3a$ with its centre at the origin?
2. Two charges of $+10\ \mu\text{C}$ and $+40\ \mu\text{C}$ respectively are placed 12 cm apart. Find the position of the point where the electric field is zero.
3. Two identical metallic spheres, having unequal, opposite charges are placed at a distance 0.90 m apart in air. After bringing them in contact with each other, they are placed at the same distance apart. Now the force of repulsion between them is 0.025 N. Calculate the final charge on each of them.
4. Three-point charges of $+2\ \mu\text{C}$, $-3\ \mu\text{C}$ and $-3\ \mu\text{C}$ are kept at the vertices A, B and C respectively of an equilateral triangle of side 20 cm as shown in figure. What should the sign and magnitude of charge to be placed at the midpoint M of side BC be so that charge at A remains in equilibrium?

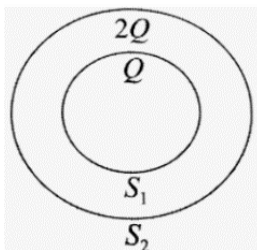


5. The electric field components due to a charge inside a cube of side 0.1 m are as shown:
 $E_x = \alpha x$, where, $\alpha = 500\ \text{N/C-m}$, $E_y = 0$, $E_z = 0$
Calculate (i) the flux through the cube and (ii) the charge inside the cube.

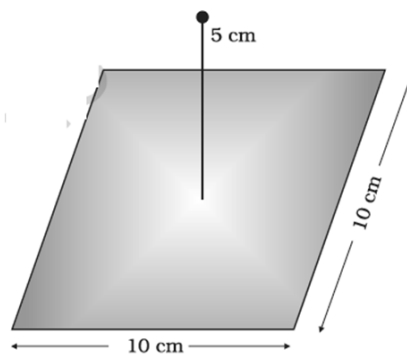


6. An electric dipole, when held at 30° with respect to a uniform electric field of $10^4\ \text{N/C}$ experiences a torque of $9 \times 10^{-26}\ \text{Nm}$. Calculate the dipole moment of the dipole.

7. A charge of $17.7 \times 10^{-4} \text{ C}$ is distributed uniformly over a large sheet of area 200 m^2 . Calculate the electric field intensity at a distance 20 cm from it in air.
8. S_1 and S_2 are two concentric spheres enclosing charges Q and $2Q$ respectively as shown in the figure.



- What is the ratio of the flux through S_1 and S_2 ?
 - How will the electric flux through the sphere S_1 change, if a medium of dielectric constant k is introduced in the space inside S_1 in place of air?
 - How will the electric flux through the sphere S_1 change, if a medium of dielectric constant k is introduced in the space inside S_2 in place of air?
9. The sum of two-point charges is $7 \mu\text{C}$. They repel each other with a force of 1 N when kept 30 cm apart in free space. Calculate the value of each charge.
10. Calculate the distance between two protons such that the electrical repulsive force between them is equal to the weight of either.
11. A point charge $+10 \mu\text{C}$ is at a distance 5 cm directly above the centre of a square of side 10 cm as shown in the figure. What is the magnitude of the electric flux through the square?



12. A point charge is placed at the centre of a spherical Gaussian surface. How will the electric flux ϕ change if
- the sphere is replaced by a cube of same or different volume,
 - a second charge is placed near and outside, the original sphere,
 - a second charge is placed inside the sphere
 - the original charge is replaced by an electric dipole?
13. Two similar balls each having mass m and charge q are hung from a silk thread of length l . Prove that the equilibrium separation

$$X = \left(\frac{q^2 l}{2\pi\epsilon_0 m g} \right)^{1/3}$$

when each thread makes a small angle θ with the vertical.

14. Two-point charges, q_1 and q_2 are located at points $(a, 0, 0)$ and $(0, b, 0)$ respectively. Find the electric field, due to both these charges, at the point $(0, 0, c)$.
15. (a) An infinitely long positively charged straight wire has a linear charge density $\lambda \text{ Cm}^{-1}$. An electron is revolving around the wire as its centre with a constant velocity in a circular plane perpendicular to the wire. Deduce the expression for its kinetic energy.
(b) Plot a graph of the kinetic energy as a function of the charge density λ .

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